Round 15 | Deep kurtosis pattern

One cheap readout, tested against stored source captures
Astra research note | 7 October 2026

I cannot give you an estimator demonstrated to meet the requested gate. I tested a distinct readout of the chain's long source history. Even a generous per-layer oracle fit reaches only **0.482 at L14 and 0.456 at L15 [Q-test]**, below 0.7 [Q]. Its cost is negligible; accuracy is the reason to reject it.

Labels: [D] means derived; [Q] means quoted from your prompt or project; [Q-test] means quoted from this run's embedded logs; [E] would mean estimated. Algebraic constants, dimensions and indices in equations are [D]. No predicted correlation is presented as a measurement.

The single candidate: retain the signed diagonal by birth

Let B_{bi} be the contribution of birth b to the current pre-activation third-cumulant diagonal. Keep the birth index in the chain's diagonal contractions instead of immediately summing it out. The temporary array B has shape k by n [D]; the chain already carries the underlying dense legs or shared-basis factors.

```
B_bi = sum_j [X1_bij P_bij Y1_bij + M_bij P_bij^2
              + X3_bij P_bij y_bi]
D3_i = sum_b B_bi;     E_i = sum_b B_bi^2 / v_i^3
```

These are the actual source blocks used in `r7_srcgain.py`: X1 = 3A + Xf; Y1 = A diag(w2) + Yf; M = P diag(s) + 3A diag(e) + ZL^T; X3 = A diag(c1) + P diag(c2). The feedback factors Xf, Yf and the transported thin vector y are retained. This is not a readout of P alone.

All births and leg transports remain the chain's existing ones. In the row-vector weight convention, an old leg H advances as H' = W_l^T diag(Phi(alpha$_l$)) H; the newborn uses its normal chain birth and raw next linear map. No new transport or independent cumulant state is introduced.

Readout and mean closure

Let x_i be your existing feature vector, including its intercept. Fit shared coefficients beta and theta once on training data, then freeze them. The proposed extension is:

```
r_i = beta.T x_i + theta E_i;      q_i = v_i^2 r_i
k4hat_i = q_i * mean(k4chain) / mean(q)
```

The mean is over all neurons, preserving the **raw diagonal mean level**, as in the current V55 implementation. Replace the diagonal in the mean closure only. Its linear Edgeworth change is phi(alpha)(alpha^2 - 1)(k4hat - k4chain)/(24 sigma^3) [D].

Why the old kill did not already test this readout

The Q22 pre-check fits powers of P with per-source scalar weights [Q]. E also uses A and the feedback/thin legs, with the source signs combined before squaring. Contributions a and -a have zero summed skew but energy 2a^2 [D]. Thus E is not determined by the summed-skew feature. It sees the full retained cubic history, including past all-distinct cubic entries mixed into the current diagonal. **That does not establish the missing fourth-order sectors.** The test below asks whether this additional information predicts them.

Exact marginal cost

n = 1024 [Q]. Scalar arithmetic is float32; one u = 2^{31} billed FLOPs [Q].

The implementation change is local to `_dslices`. Its diagonal consists of four contractions, schematically 3a + b + 3c + d [D]. Change their output index from i to bi, combine them into B, then sum B over births to recover the same D3. The existing matrices and thin contractions are reused.

Incremental operation	Exact extra scalar FLOPs [D]	Extra primitive calls [D]
Retain birth index; sum B at the end	2n(k - 1)	1
Compute E: square B, sum; form v*v*v; divide	2kn + 2n	5
Add theta E to the feature prediction	2n	2
Total per corrected layer	4kn + 2n	8

Keeping the birth index does not add matrix products. In the first row, the reduction additions merely move to the final birth sum; the extra arithmetic is the two coefficient scalings now applied per birth. This ledger explicitly forms v*v*v by multiplication, rather than a generic power operator. Existing level rescaling is shared.

Meter check at the actual width

Sources k	Original kernel [Q-test]	Extended kernel [Q-test]	Difference [Q-test]
7	51,395,584 FLOPs 9 operations	51,426,304 FLOPs 17 operations	30,720 FLOPs +8 operations
15	110,132,224 FLOPs 9 operations	110,195,712 FLOPs 17 operations	63,488 FLOPs +8 operations

The installed meter is flopscope 0.12.1 [Q-test]. These are isolated-kernel measurements with ordinary float32 arrays at the real shapes, not a full graded rollout. Changing the accumulation order produced about 10^{-7} relative differences in the checked diagonals [Q-test].

Whole-network increment

```
Correct layers 8..16: k = 7..15; sum k = 99
Delta FLOPs = 4*1024*99 + 2*1024*9 = 423,936
            = 0.0001974105835 u
Delta kernel operations = 8*9 = 72; dense products = 0
```

All whole-network quantities above are **[D]**, for the stated integration. B occupies at most 15 by 1024 float32 values, or 60 KiB [D], plus scratch buffers. There is no need to retain B between layers: the existing source factors already retain the history.

The kernel fits comfortably below the 20-30 u and 1,000-extra-operation allowances [Q]. Buffer setup and surrounding integration were not metered in a full estimator; the 72-call result is the incremental kernel ledger, not a claim about a submitted grader trace. I did not edit the submission or run a coupled mean rollout.

What the low cost does not buy

This is a nonlinear readout of a rich cubic state. It is not an exact fourth-cumulant closure. Long history and sign-sensitive source factors are necessary information to examine, but they do not imply correlation at the requested level. No theorem or result in the prompt supplies that missing implication.

One falsifier, executed

Public MLP #0 | stored dense source captures | teacher used only to score and form oracle bounds

Test: reconstruct B at the captured layers, append E to the existing feature span, then give the candidate arbitrary coefficients separately at each layer. For the correlation bound, additionally allow a constant in teacher-standardized units and remove the level constraint. Least squares on all decisive neurons then gives the maximum correlation in this enlarged linear family. Any shared-coefficient candidate above is contained in it.

Targets are teacher excess kurtosis $K4d/v_{true}^2$ on $|mu_{true}/sigma_{true}| < 1.5$ [Q]. Source sums reproduce the stored chain D3 to at most 1.34×10^{-7} relative error [Q-test]. The table's baseline is the reconstructed feature span on these older dense captures; it is not a rerun of the current graded model.

Layer	E alone: correlation	Baseline: best correlation	+ E: best correlation	Best weighted norm: (+ E) / baseline
7	0.125	0.581	0.581	0.9995
11	0.032	0.526	0.527	1.0000
14	0.021	0.478	0.482	0.9967
15	0.032	0.455	0.456	0.9921

All numerical table entries are [Q-test]. The last column uses separate optimal weighted fits for each feature family, over all neurons. It is a ratio of error norms, not squared errors. The correlation and weighted-error optima need not use the same coefficients. Both are deliberately optimistic diagnostics.

The blend gate also fails in this diagnostic

```
G4_i = phi(alpha_i) (alpha_i^2 - 1)/(24 sigma_i^3)
err(k) = || G4 * (k - k4true) ||_2
```

Even dropping the level constraint, the best extended-family errors at L14/L15 are 0.893/0.902 times the original stored chain's error [Q-test]. Against the best baseline feature fit they are 0.9967/0.9921 [Q-test]. For comparison, a literal half blend with the true diagonal has error norm ratio 0.5, or squared-error ratio 0.25 [D]. Correlation 0.7 alone is not mathematically equivalent to that blend criterion [D].

A less generous check fits on L7/L11 and applies the frozen coefficients at L14/L15: correlations are 0.359/0.329 [Q-test], after preserving the chain's raw mean level. Thus the failure is not caused by insisting on cross-layer transfer; it persists even under the oracle fit.

Decision and scope

Reject this readout. It barely adds information to the feature baseline on the stored state. Failure at the tested deep layers suffices to reject the all-layer gate for this candidate on those captures. L12/L13/L16 source readouts, the synthetic network and a current-chain coupled rollout were not tested. This does not rule out every nonlinear readout of the full source factors, and it does not identify a qualifying estimator.

Reproduce: the PDF embeds the three Python scripts and their three JSON logs. Run `probe_birth_energy.py`, then `audit_ceiling.py`; use the project virtual environment for `meter_birth_energy.py`. The scripts read the project and write beside themselves. Sources: your Round 15 prompt; `experiments/probes/r7_srcgain.py`, `archA_q22_precheck.py`; `submissions/v55lean/estimator.py`, especially the mean-level rescaling and `_dslices`.